

Module 07: Communication — Robot Talk

Learning Objectives

1. Understand that robots speak a secret language called **Binary** (1s and 0s).
2. Learn about **Wireless** signals (Invisible waves like Wi-Fi).
3. Know that robots can talk to each other to work as a team.

Introduction: The Secret Code

Do robots speak English? Not really. Inside their brains, they speak **Math**. They only use two numbers: **0** and **1**.

- 0 = OFF (No electricity).
- 1 = ON (Electricity!). This is called **Binary Code**. Every letter you type is turned into 0s and 1s for the computer to understand!

Key Concepts

1. Binary (The Language of Machines)

A = 01000001 B = 01000010 C = 01000011 It looks crazy to us, but for a robot, it is very clear!

2. Wireless Signals (Invisible Wires)

How does a drone fly without a wire? It uses **Radio Waves** (like Wi-Fi or Bluetooth). The controller sends invisible messages through the air.

- "Go Up!" -> *Zap* -> Robot goes up.

3. Swarm Intelligence (Teamwork)

Ants talk to each other to find food. Robots can do this too! A "Swarm" of drones can fly together without crashing. They are constantly sending messages: "I am here! Move left!"

Activities

Activity 1: Binary Beep

Play a game with a flashlight.

- Flash = 1.
- No Flash = 0. Send a code to your friend. Can they decode it?

Activity 2: The Robot Swarm

3 friends are robots. Hold a long string between you (The Network). Close your eyes. Feel the string pull. If it pulls Left -> Everyone move Left. You are communicating through the web!

Summary

Communication allows robots to share information. Whether through wires or invisible waves, they use the language of Binary (On/Off) to talk to the world.

References

- *How to Talk to Your Computer* by Seymour Simon
- *Secret Coders* by Gene Luen Yang